1 A thin copper sheet is suspended from a small hole near the top of the sheet and placed in a magnetic field, as shown in Fig. 1.1.

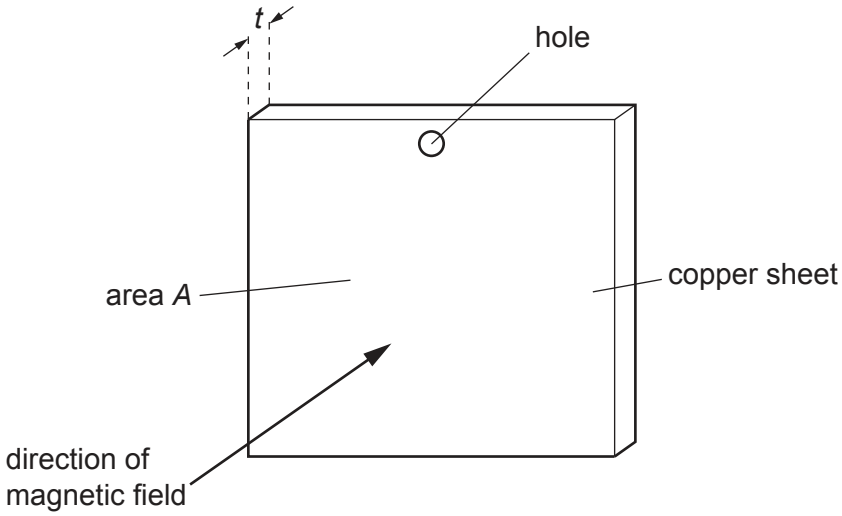

Fig. 1.1 (not to scale)

The sheet has area A and thickness t.

The sheet is displaced from its equilibrium position through a horizontal distance s_0 and then released so that it oscillates perpendicular to the direction of the magnetic field. The horizontal distance s of the sheet from its equilibrium position is measured after five complete oscillations.

It is suggested that s is related to A by the relationship

$$s = s_0 e^{-ABKt}$$

where B is the magnetic flux density of the field and K is a constant.

Plan a laboratory experiment to test the relationship between s and A.

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine a value for K.

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

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